

FERC Order Approving IBR and Generators Modeling Reliability Standards (2026)

FERC ORDER APPROVING INVERTER-BASED RESOURCES AND GENERATORS MODELING RELIABILITY STANDARDS

Docket Nos. RD26-1-000, RD26-2-000, RD26-3-000

Issued February 19, 2026

1. In Order No. 901, the Commission directed the North American Electric Reliability Corporation (NERC), the Commission-certified Electric Reliability Organization (ERO), to develop new or modified Reliability Standards that address specific matters pertaining to the impacts of inverter-based resources (IBR) on the reliable operation of the Bulk-Power System. Due to the significant scope of the work, the Commission required NERC to submit responsive Reliability Standards in three tranches. On November 4, 2025, NERC submitted three petitions seeking approval of five proposed Reliability Standards and related definitions, representing the second tranche of Reliability Standards directed by the Commission in Order No. 901.

2. Specifically, in the first petition (RD26-1-000), NERC seeks approval of: (1) a proposed definition of the term distributed energy resource (DER) for inclusion in the NERC Glossary of Terms Used in NERC Reliability Standards (NERC Glossary); and (2) proposed Reliability Standards MOD-032-2 (Data for Power System Modeling and Analysis), IRO-010-6 (Reliability Coordinator Data and Information Specification and Collection), and TOP-003-8 (Transmission Operator and Balancing Authority Data and Information Specification and Collection). The second petition (RD26-2-000) seeks approval of Reliability Standard MOD-033-3 (Steady-State and Dynamic System Model Validation). The third petition (RD26-3-000) seeks approval of: (1) proposed definitions of model validation and model verification in the NERC Glossary; and (2) proposed Reliability Standard MOD-026-2 (Verification and Validation of Dynamic Models and Data).

3. For the reasons discussed below, pursuant to section 215(d)(2) of the Federal Power Act (FPA), we grant the requested approvals. Our action approving the three petitions and the associated Reliability Standards should ensure that Bulk-Power System planners and operators will have the data and models needed to plan for, operate, and reliably integrate IBRs on the Bulk-Power System.

Proposed Definitions:

DER Definition: NERC proposes to define DER as: "A generator or energy storage technology connected to a distribution system that is capable of providing Real Power in non-isolated parallel operation with the Bulk-Power System, including one connected behind the meter of an end-use customer that is supplied from a distribution system."

Model Validation Definition: "The process of comparing simulation results with measurements to assess how closely a model's behavior matches the measured behavior."

Model Verification Definition: "The process of confirming that model structure and parameter values are representative of the equipment or facility design and settings by reviewing equipment or facility design and settings documentation."

Proposed Reliability Standard MOD-032-2:

NERC states that the purpose of proposed Reliability Standard MOD-032-2 is "to establish consistent modeling data requirements and reporting procedures for development of planning horizon cases necessary to support analysis of the reliability of the interconnected transmission system." The proposed Standard requires planning coordinators, working jointly with their associated transmission planner(s), to develop steady-state, dynamic, and short circuit modeling data requirements and reporting procedures.

Proposed Reliability Standards IRO-010-6 and TOP-003-8:

NERC states that the purpose of proposed Reliability Standard IRO-010-6 is "to prevent instability, uncontrolled separation, or cascading outages that adversely impact reliability, by ensuring each reliability coordinator has the data and information it needs to plan, monitor and assess the operation of its reliability coordinator area." NERC proposes minor modifications to proposed Reliability Standards IRO-010-6 and TOP-003-8 to add "Inverter-based Resource (IBR)-specific data and parameters" to the list of information reliability coordinators, transmission operators, and balancing authorities must address in their documented data specifications.

Proposed Reliability Standard MOD-033-3:

NERC states that the purpose of proposed Reliability Standard MOD-033-3 is "to establish a process for system model validation to facilitate achieving and maintaining model accuracy." The proposed Standard requires each planning coordinator to develop and implement a model validation process for the planning system models developed in accordance with Reliability Standard MOD-032, representing its portion of the existing system. The model validation process includes the comparison of power flow simulation performance of the steady-state planning system model to actual system behavior at least once every 24 calendar months.

Proposed Reliability Standard MOD-026-2:

NERC states that the purpose of proposed Reliability Standard MOD-026-2 is "to verify and validate that the dynamic models and associated parameters used to assess Bulk Electric System (BES) reliability represent the in-service equipment of Bulk Power System facilities including generating facilities, transmission connected dynamic reactive resources, and high-voltage direct current (HVDC) systems." Reliability Standard MOD-026-2 consolidates the currently effective Reliability Standards MOD-026-1 and MOD-027-1. The proposed Standard requires each generator owner or transmission owner to provide its transmission planner positive sequence dynamic model(s) with associated parameters, any information pertaining to changes to the model(s) or its parameters, and accompanying documentation. Requirement R3 of the proposed Standard requires each generator owner or transmission owner to provide EMT models for flexible alternating current transmission system (FACTS) devices, HVDC systems, and registered IBRs to its transmission planner. Legacy facilities are excluded from this requirement where the original equipment manufacturer no longer supports EMT model(s) for the facility.

Determination:

Pursuant to section 215(d)(2) of the FPA, we approve the proposed DER definition for inclusion in the NERC Glossary, as well as proposed Reliability Standards MOD-032-2, IRO-010-6 and TOP-003-8, as just, reasonable, not unduly discriminatory or preferential, and in the public interest. We find that proposed Reliability Standards MOD-032-2, IRO-010-06, and TOP-003-8 are responsive to the relevant directives in

Order No. 901 and improve upon the existing standards by supporting accurate modeling of IBRs, which will advance the reliability of the Bulk-Power System.

We also approve proposed Reliability Standard MOD-033-3, as just, reasonable, not unduly discriminatory or preferential, and in the public interest, finding it responsive to relevant directives in Order No. 901 and an improvement upon the existing standard by requiring the inclusion of IBRs in system-wide models.

We approve the proposed model verification and model validation definitions for inclusion in the NERC Glossary, as well as proposed Reliability Standard MOD-026-2, finding it responsive to the relevant directives in Order No. 901 and an improvement upon existing Reliability Standards MOD-026-1 and MOD-027-1 by establishing a process for the model validation and model verification of IBRs.

By the Commission.

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Debbie-Anne A. Reese, Secretary.